

Syntax-Based Belief Merging with Unknown Reliability - Supplementary Materials

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1 Introduction

In this supplementary materials file, the first section is dedicated to the proofs of the properties satisfied by our methods. The second section provides the proofs of the propositions related to our metrics.

2 Proofs for the Logical Properties IC

In this section, we study the theoretical properties of our new family of operators.

Definition 1. Let E be a profile, let IC be a formula coding the integrity constraints of the merging, and let Δ be an operator that assigns to each profile E and formula IC a knowledge base $\Delta_{IC}(E)$. Δ is an IC merging operator if and only if it satisfies:

(IC0) $\Delta_{IC}(E) \vdash IC$

(IC1) If IC is consistent then $\Delta_{IC}(E)$ is consistent

(IC2) If $\bigwedge E$ is consistent with IC , then $\Delta_{IC}(E) \equiv \bigwedge E \wedge IC$

(IC3) If $E_1 \leftrightarrow E_2$ and $IC_1 \leftrightarrow IC_2$, then $\Delta_{IC_1}(E_1) \leftrightarrow \Delta_{IC_2}(E_2)$

(IC4) If $K \vdash IC$ and $K' \vdash IC$, then $\Delta_{IC}(K \sqcup K') \wedge K \not\vdash \perp \Rightarrow \Delta_{IC}(K \sqcup K') \wedge K' \not\vdash \perp$

(IC5) $\Delta_{IC}(E_1) \wedge \Delta_{IC}(E_2) \vdash \Delta_{IC}(E_1 \sqcup E_2)$

(IC6) If $\Delta_{IC}(E_1) \wedge \Delta_{IC}(E_2)$ is consistent, then $\Delta_{IC}(E_1 \sqcup E_2) \vdash \Delta_{IC}(E_1) \wedge \Delta_{IC}(E_2)$

(IC7) $\Delta_{IC_1}(E) \wedge \Delta_{IC_2}(E) \vdash \Delta_{IC_1 \wedge IC_2}(E)$

(IC8) If $\Delta_{IC_1}(E) \wedge \Delta_{IC_2}(E)$ is consistent then, $\Delta_{IC_1 \wedge IC_2}(E) \vdash \Delta_{IC_1}(E)$

(IC0) states that the result of merging satisfies the integrity constraints. (IC1) ensures that, when the integrity constraints are consistent, we always manage to extract a coherent piece of information from the profile. (IC2) says that, if possible, the result of merging is simply the conjunction of the knowledge bases with the integrity constraints. (IC3) is the principle of irrelevance of syntax: the result of merging has to depend only on the expressed opinions and not on their syntactical presentation. (IC4) is a fairness postulate meaning that the result of merging of two knowledge bases should not give preference to one of them, if it is consistent with one of them, it should be consistent with the other one. (IC5) expresses the following idea: if belief profiles are viewed as expressing the beliefs of the members of a group, then if E_1 (corresponding to the first group) compromises

on a set of alternatives which A belongs to, and E_2 (corresponding to the second group) compromises on another set of alternatives which contains A too, then A has to be in the chosen alternatives if we join the two groups. (IC5) and (IC6) together state that if one could find two subgroups which agree on at least one alternative, then the result of the global merging will be exactly those alternatives the two groups agree on.

Let us check what are the properties satisfied by our operators.

Proposition 1. Whatever the aggregation function $\oplus$, the merging operator $\Delta_{IC}^{R\oplus}$ satisfy (IC0), (IC1), (IC2), (IC7) and (IC8), but does not satisfy (IC3), (IC4), (IC5) and (IC6).

Proof. (IC0) and (IC1) are satisfied by definition.

(IC2) is satisfied. When $\bigwedge E$ is consistent with IC , this means that the conjunction of the formulae supported in the extended Uprofile $\underline{\Psi}$ is a maxcons consistent with IC . Then the result of the merging with the operator $\Delta_{IC}^{R\oplus}(E)$ will be this maxcons because it will maximize the reliability (because of the non-decreasingness of $\oplus$).

(IC3) is not satisfied. Consider $IC = \top$, $K_1 = \{a, b\}$, $K_2 = \{a \wedge b\}$ and $K_3 = \{\neg b\}$. Let $E_1 = K_1 \sqcup K_3$ and $E_2 = K_2 \sqcup K_3$ be the two equivalent profiles.

We have the extended Uprofiles $\underline{\Psi}_1 = \{a, \neg a, b, \neg b\}$ and $\underline{\Psi}_2 = \{b, \neg b, a \wedge b, \neg(a \wedge b)\}$. The reliabilities are computed and reported in the following tables:

	K_1	K_3	a	$\neg a$	b	$\neg b$
It1	1	1	1	0	1	1
It2	0.75	0.25	0.75	0	0.75	0.25
It3	1	0	1	0	1	0

Table 1: Reliability of E_1

	K_2	K_3	b	$\neg b$	$a \wedge b$	$\neg(a \wedge b)$
It1	1	1	1	1	1	1
It2	0.5	0.5	0.5	0.5	0.5	0.5

Table 2: Reliability of E_2

Whatever the aggregation function $\oplus$, the results of the merging are $\Delta_{\top}^{R\oplus}(E_1) = \{a, b\}$ and $\Delta_{\top}^{R\oplus}(E_2) =$

$\{\{-b, \neg(a \wedge b)\}, \{b, \neg(a \wedge b)\}, \{b, a \wedge b\}\}$. Each maxcons of $\Delta_{\top}^{R\oplus}(E_1)$ implies a , but this is not the case with the maxcons of $\Delta_{\top}^{R\oplus}(E_2)$. Even though E_1 and E_2 are logically equivalent, the result of the merging of E_1 implies a but this will not be the case with the merging of E_2 .

(IC4) is not satisfied. Consider $K = \{a, b, a \wedge b\}$, $K' = \{-a\}$ and $IC = \top$. The reliability of the agents and the formulae are in Table 3 and Table 4.

	K	K'
It1	1	1
It2	0.833	0.167
It3	1	0

Table 3: Reliability of the agents

	a	$\neg a$	b	$\neg b$	$a \wedge b$	$\neg(a \wedge b)$
It1	1	1	1	0	1	0
It2	0.833	0.167	0.833	0	0.833	0
It3	1	0	1	0	1	0

Table 4: Confidence of the formulae

Whatever the aggregation function $\oplus$, the maxcons that maximizes the confidence of the formulae is $\{a, b, a \wedge b\}$ with a confidence of 1 and $\{a, b, a \wedge b\} \wedge K' \vdash \perp$ whereas it is consistent with K (equivalent, in fact).

(IC5) and (IC6) are not satisfied. Consider $\phi_1 = (a \wedge b) \vee (\neg a \wedge \neg b \wedge c)$, $\phi_2 = \neg a \wedge b$ and $\phi_3 = (a \wedge \neg c) \vee \neg b$.

Consider the bases $K_1 = K_2 = K_3 = \{\phi_1, \neg\phi_2\}$, $K_4 = K_5 = \{\neg\phi_1, \phi_2\}$, $K_6 = K_7 = \{\phi_2, \neg\phi_3\}$ and $K_8 = K_9 = K_{10} = \{\neg\phi_2, \phi_3\}$.

Let $E_1 = \{K_1, K_2, K_3, K_4, K_5\}$, $E_2 = \{K_6, K_7, K_8, K_9, K_{10}\}$ be the profiles.

When we build the profiles, the bases K_i and the formulae supported s_{K_i} are the same with E_1 and E_2 but with the union of the two profiles, we have some adjustments to do for some agents because they support a new formula with the union of profiles. With $E_1 \sqcup E_2$ we have $s_{K_4} = s_{K_5} = s_{K_6} = s_{K_7} = \{\neg\phi_1, \phi_2, \neg\phi_3\}$. For the other agents, we have $s_{K_i} = K_i$ with $i \in \{1, 2, 3, 8, 9, 10\}$.

The reliability of the agents and the formulae are in Table 5 for E_1 , Table 6 for E_2 and Table 7 for $E_1 \sqcup E_2$.

After the merging, whatever the aggregation function $\oplus$, we have $\Delta_{\top}^{R\oplus}(E_1) = \{\{\phi_1, \neg\phi_2\}\}$, $\Delta_{\top}^{R\oplus}(E_2) = \{\{\neg\phi_2, \phi_3\}\}$ and $\Delta_{\top}^{R\oplus}(E_1 \sqcup E_2) = \{\{\neg\phi_1, \phi_2, \neg\phi_3\}\}$. So (IC5) and (IC6) are not satisfied.

(IC7) and (IC8) are satisfied. The reliability of the agents is the same with $\Delta_{IC_1}^{R\oplus}(E)$ and $\Delta_{IC_1 \wedge IC_2}^{R\oplus}(E)$ because the profile E does not change. The only change is in the generation of the maxcons.

The case where $\Delta_{IC_1}^{R\oplus}(E) \wedge IC_2$ is not consistent is trivial.

Suppose that $\Delta_{IC_1}^{R\oplus}(E) \wedge IC_2$ is consistent. Let $M \in \text{MAXCONS}(E, IC_1)$ s.t. $M \in \Delta_{IC_1}^{R\oplus}(E)$ and M is consistent with IC_2 . As M is consistent with $IC_1 \wedge IC_2$, it is clear that $M \in \text{MAXCONS}(E, IC_1 \wedge IC_2)$. Furthermore, the reliability of M is computed from the same profile

E and then if M is selected by $\Delta_{IC_1}^{R\oplus}(E)$, it is also the case for $\Delta_{IC_1 \wedge IC_2}^{R\oplus}(E)$. (IC7) is satisfied.

Let $M \in \Delta_{IC_1 \wedge IC_2}^{R\oplus}(E)$. Then M is also a maxcons of $\text{MAXCONS}(E, IC_1)$ with a maximal reliability, which is selected by $\Delta_{IC_1}^{R\oplus}(E)$: (IC8) is satisfied.

	$K_{\{1,2,3\}}$	$K_{\{4,5\}}$	ϕ_1	$\neg\phi_1$	ϕ_2	$\neg\phi_2$
It1	1	1	3	2	2	3
It2	1	0	3	0	0	3

Table 5: Reliability of E_1

	$K_{\{6,7\}}$	$K_{\{8,9,10\}}$	ϕ_2	$\neg\phi_2$	ϕ_3	$\neg\phi_3$
It1	1	1	2	3	3	2
It2	0	1	0	3	3	0

Table 6: Reliability of E_2

	$K_{\{1,2,3\}}$	$K_{\{4,5,6,7\}}$	$K_{\{8,9,10\}}$	ϕ_1	$\neg\phi_1$	ϕ_2	$\neg\phi_2$	ϕ_3	$\neg\phi_3$
It1	1	1	1	3	4	4	6	3	4
It2	0.33	0.66	0.33	1	2.66	2.66	2	1	2.66
It3	0	1	0	0	4	4	0	0	4

Table 7: Reliability of $E_1 \sqcup E_2$

□

The results are summarized in Table 8 together with the properties satisfied by standard syntax-based merging operators (those other results come from (Konieczny 2000)).

3 Truth-tracking Metrics

In this section, we provide the proofs of the propositions that examine the relationships between the three metrics used.

Proposition 2. *For any operator and any set $\mathcal{M}$, we have $\text{str_prec}(\mathcal{M}) \leq \text{avg_prec}(\mathcal{M})$.*

Proof. If $\mathcal{M}$ is s.t. $\text{str_prec}(\mathcal{M}) = 0$, $\text{str_prec}(\mathcal{M}) \leq \text{avg_prec}(\mathcal{M})$ because $\text{avg_prec}(\mathcal{M}) \geq 0$.

If $\mathcal{M}$ is s.t. $\text{str_prec}(\mathcal{M}) \neq 0$, it means that $T \in \mathcal{M}$ and $\text{str_prec}(\mathcal{M}) = \frac{1}{|\mathcal{M}|}$.

$$\text{avg_prec}(\mathcal{M}) = \frac{\sum_{M \in \mathcal{M}} \frac{|M \cap T|}{|M|}}{|\mathcal{M}|}$$

$$\text{avg_prec}(\mathcal{M}) = \frac{\sum_{M \in \mathcal{M}, M \neq T} \frac{|M \cap T|}{|M|}}{|\mathcal{M}|} + \frac{\frac{|T \cap T|}{|T|}}{|\mathcal{M}|}$$

As $\frac{|T \cap T|}{|T|} = \frac{1}{|T|}$, again, $\text{str_prec}(\mathcal{M}) \leq \text{avg_prec}(\mathcal{M})$ and the result holds. □

Proposition 3. *For any operator and any set $\mathcal{M}$, the results of the metrics $\text{una_prec}(\mathcal{M})$ and $\text{avg_prec}(\mathcal{M})$ are independent.*

Proof. Consider the two maxcons as the result of the merging i.e. $\mathcal{M} = \{M_1, M_2\}$ with $M_1 = \{\phi_1, \phi_2\}$ and $M_2 = \{\neg\phi_1, \neg\phi_2, \phi_3, \phi_4, \phi_5\}$.

	(IC0)	(IC1)	(IC2)	(IC3)	(IC4)	(IC5)	(IC6)	(IC7)	(IC8)
$\Delta_{IC}^{R_{\oplus}}(E)$	✓	✓	✓	✗	✗	✗	✗	✓	✓
$\Delta_{IC}^{C^4}(E)$	✓	✓	✓	✗	✗	✗	✗	✓	✓
$\Delta_{IC}^D(E)$	✓	✓	✓	✗	✓	✓	✗	✓	✗
$\Delta_{IC}^{S,\Sigma}(E)$	✓	✓	✓	✗	✓	✗	✗	✓	✓
$\Delta_{IC}^{\cap,\Sigma}(E)$	✓	✓	✓	✗	✗	✓	✓	✓	✓

Table 8: Satisfaction of *IC* postulates

ω^* is model of the formulae $\neg\phi_1, \neg\phi_2, \phi_3, \phi_4$ and ϕ_5 .

We have $U(\mathcal{M}) = \{\phi_3, \phi_4, \phi_5\}$.

$una_prec(\mathcal{M}) = \frac{3}{5}$ and $avg_prec(\mathcal{M}) = \frac{\frac{0}{2} + \frac{5}{5}}{2} = \frac{1}{2}$, so $una_prec(\mathcal{M}) > avg_prec(\mathcal{M})$.

But with $M_1 = \{\phi_1, \phi_2, \phi_3\}$ and $M_2 = \{\neg\phi_1, \neg\phi_2, \neg\phi_3\}$, we have $U(\mathcal{M}) = \emptyset$ and so $avg_prec(\mathcal{M}) > una_prec(\mathcal{M})$.

The results of the two metrics are independent. $\square$

Proposition 4. *For any operator and any set $\mathcal{M}$, the results of the metrics $str_prec(\mathcal{M})$ and $una_prec(\mathcal{M})$ are independent.*

Proof. Consider the two maxcons as the result of the merging i.e. $\mathcal{M} = \{M_1, M_2\}$ with $M_1 = \{\phi_1, \phi_2, \phi_3\}$ and $M_2 = \{\neg\phi_1, \neg\phi_2, \neg\phi_3\}$. ω^* is in the set of interpretations of the formulae ϕ_1, ϕ_2 and ϕ_3 . We have $U(\mathcal{M}) = \emptyset$. $str_prec(\mathcal{M}) = \frac{1}{2}$ and $una_prec(\mathcal{M}) = 0$, so $str_prec(\mathcal{M}) > una_prec(\mathcal{M})$.

But if we change M_2 with $M_2 = \{\phi_1, \phi_2, \neg\phi_3\}$, we have $una_prec(\mathcal{M}) = \frac{2}{3}$ and $str_prec(\mathcal{M}) < una_prec(\mathcal{M})$.

The results of the two metrics are independent. $\square$

References

Konieczny, S. 2000. On the difference between merging knowledge bases and combining them. In *Proceedings of the 7th International Conference on Principles of Knowledge Representation and Reasoning (KR 2000)*, 135–144.